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Roll No

EE-701-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Power System Analysis and Control

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Derive the expression for steady state frequency change for single area system with following cases
 - i) Change in load with fixed speed
 - ii) Change in speed with fixed demand
2. What is power system stability? Clarify the types of stability. Give the methods to improve transient stability.
3.
 - a) Discuss the reasons for keeping strict limit on the system frequency variations.
 - b) Two turbo alternators rated 110 mw and 210 mw have governor drop characteristics of 5 percent from no load to full load. They are connected in parallel to share a load of 250 mw. Determine the load shared by each machine assuming per governor action.
4.
 - a) With the help of suitable diagram explain the working of an automatic voltage regulator.
 - b) Discuss the utility of excitation control.

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5.
 - a) Describe the speed governing system.
 - b) Draw the block diagram representation of single area interconnected system.

6.
 - a) Derive the swing equation of a synchronous machine.
 - b) A two pole 50Hz 60MVA turbogenerator has a moment of inertia of $9 \times 10^3 \text{ kg-m}^2$. Calculate:
 - i) The kinetic energy in mJ at rated speed
 - ii) The inertia constants M and H
 - iii) The inertia constant on 50MVA base

7.
 - a) Explain the concept of power system voltage stability and give the definition of voltage stability.
 - b) Derive swing equation and discuss its application in the study of power system stability.

8. Write short notes (Any two):
 - a) Deregulation of power system
 - b) Economic Dispatch
 - c) Comparison of different power flow studies method
 - d) Load Damping
